LOGARITHM

ALGEBRA

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

70%

WEEKLY

30%

EXAM

WEEKLY GRADE

70%

LONG QUIZ

30%

SEATWORK

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Logarithmic Equation

Natural Logarithm

- Napier's number, e

Properties of Logarithm

- Product, Quotient, Power Rule
- Change of Base Rule

LOGARITHMIC EQUATION

Credited as the inventor of logarithms

John Napier

DEFINITION OF LOGARITHM

"2 raised to which power equals 16?"

Logarithms are another way of thinking about **exponents**

$$2 \cdot 2 \cdot 2 \cdot 2 = 16$$

$$\mathbf{2^4} = 16$$

Exponential Form

$$b^c = a$$

where,

b is the base

c is the exponent

a is the argument

Logarithmic Form

$$\log_b a = c$$

$2^3 = 8$ $\quad$ $log_2 8 = 3$

$3^4 = 81$ $\quad$ $log_3 81 = 4$

$5^2 = 25$ $\quad$ $log_5 25 = 2$

EXERCISE

Which of the following is **equivalent** to

$\mathbf{2^5 = 32}$?

A. $log_2 32 = 5$

B. $log_5 2 = 32$

C. $log_{32} 5 = 2$

EXERCISE

Write $\boldsymbol{log_4 16 = 2}$ in **exponential** form.

NATURAL LOGARITHM

1

$\downarrow$ 100% (1 year)

$1 + 1 = 2$

$$= 1\left(1 + \frac{1}{1}\right)^{1}$$

1

$\downarrow$ 50% (6 months)

$1 + 0.5 = 1.5$

$\downarrow$ 50% (6 months)

$1.5 + 0.75 = 2.25$

$$= 1\left(1 + \frac{1}{2}\right)^{2}$$

NAPIER'S CONSTANT, e

$$e = \lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n$$

$$e = 2.718281828 \ldots$$

THE NATURAL LOGARITHM

A logarithm with **base**, e

Exponential Form

$$e^c = a$$

Logarithmic Form

$$\log_e a = c$$

$$\ln a = c$$

Solve for x,

$$5 \cdot 2^x = 640$$

Solve for x,

$$4 = log_2 \frac{256}{x}$$

PROPERTIES OF LOGARITM

PRODUCT RULE

Expand $\log_6 5y$

$$\log_b MN = \log_b M + \log_b N$$

Condense $log_3 10 + log_3 x$

where,

$M, N > 0$ *and* $0 < b \neq 0$

QUOTIENT RULE

Expand $\boldsymbol{log_7 \frac{a}{2}}$

$$\boldsymbol{log_b \frac{M}{N} = log_b M - log_b N}$$

Condense $\boldsymbol{log_4 x^3 - log_4 y}$

where,

$M, N > 0$ *and* $0 < b \neq 0$

POWER RULE

Expand $\boldsymbol{log_7\, x^5}$

$$\boldsymbol{log_b\, M^p = p\, log_b\, M}$$

Condense $\boldsymbol{6\, ln\, y}$

where,

$M, N > 0$ *and* $0 < b \neq 0$

CHANGE OF BASE RULE

Express $\boldsymbol{log_b 4}$ as base of a

$$\boldsymbol{log_b M = \frac{log_a M}{log_a b}}$$

Express $\boldsymbol{6\ ln\ y}$

where,

$M, N > 0$ *and* $0 < b \neq 0$

Suppose there is a population of zebras in **1990** with **100** individuals. It is known that the zebra population is growing at a constant rate of **5%** per year. Now, we want to determine the **population** of zebras in the year **2020**.

The population growth formula:

$$\boldsymbol{P = P_0 e^{rt}}$$

where,

P = total population after time 't'

P_0 = starting population

r = % rate of growth

t = time

SEATWORK